



3N+1 PROBLEM 4

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The $3N+1$ sequence is generated using the following rules: If n is even, the next number is $n / 2$. If n is odd, the next number is $3 \times n + 1$. The sequence continues until it reaches 1. The cycle length of a number is the total number of terms in the sequence, including the starting number and 1.

Given two integers i and j , determine the maximum cycle length among all integers between i and j (inclusive).

If $i > j$, swap them before processing.

Input Format

A single line containing two space-separated integers:

i j

Constraints

$1 \leq i, j \leq 1,000,000$

Output Format

Print three integers:

i j maximum_cycle_length

using the original values of i and j .

Sample Input 0

1 10

Sample Output 0

1 10 20

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Language

Python 3



```
1 import sys
2
3 def get_cycle_length(n, memo):
4     if n in memo:
5         return memo[n]
6
7     if n % 2 == 0:
8         next_n = n // 2
9     else:
10        next_n = 3 * n + 1
11
12    length = 1 + get_cycle_length(next_n, memo)
13
14    if n <= 1000000:
```

```
15         memo[n] = length
16
17     return length
18
19 def solve():
20     input_data = sys.stdin.read().split()
21     if not input_data:
22         return
23
24     orig_i = int(input_data[0])
25     orig_j = int(input_data[1])
26
27     start = min(orig_i, orig_j)
28     end = max(orig_i, orig_j)
29
30     memo = [1] * 11
```

Line: 42 Col: 1



Upload Code as File



Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

Download

```
1 | 1 10
```

Your Output (stdout)

```
1 | 1 10 20
```

Expected Output

Download

```
1 | 1 10 20
```